

ALA:2

Name: Parmar Vishva j .
Enrolment No: 240905091015
Course: BET-/IT
Subject: Cloud Computing (BETCE17336)

ALA 2

Virtualization Implementation: In this activity, each student will install and configure a virtual machine or container-based environment to understand the fundamentals of virtualization. They will demonstrate resource allocation (CPU, memory, storage) and isolation between multiple virtual instances. The implementation should showcase the creation, configuration, and execution of virtual environments. Students will submit the configuration details, source/setup files, and screenshots demonstrating successful virtualization, and upload them on the GMIU portal.

Solution

1. Aim

To install and configure Ubuntu Linux as a virtual machine using Oracle VirtualBox and understand virtualization concepts, resource allocation, virtual storage, and isolation between the host and guest operating systems.

2. Software Requirements

Component	Details
Host operating system	Windows 10
Virtualization software	Oracle VirtualBox
Guest operating system	Ubuntu Linux
Processor	Multi-core CPU supporting virtualization

RAM	Minimum 4 GB recommended for the VM
Storage	Virtual hard disk, dynamically allocated or fixed-size
Network	VirtualBox NAT or Bridged Adapter

Use the actual configuration of your system when completing the report.

3. Introduction

Virtualization is a technology that allows multiple operating systems to run on a single physical computer. It uses software called a hypervisor to create and manage virtual machines.

In this activity, Oracle VirtualBox is used to run Ubuntu Linux as a guest operating system on a Windows host machine. The virtual machine has its own allocated memory, processor resources, virtual hard disk, and network adapter.

This implementation demonstrates how a virtual environment can operate independently while sharing the physical resources of the host computer.

4. Virtualization Architecture

Physical Hardware

CPU · RAM · SSD/HDD · Network Adapter

Host Operating System

Windows 10/11

Oracle VirtualBox

Virtualization software (hypervisor)

Ubuntu Linux Virtual Machine

Virtual CPU · Virtual RAM · Virtual Disk · Virtual Network

5. Implementation Steps

Step 1: Install Oracle VirtualBox

1. Download Oracle VirtualBox from the official website: virtualbox.org.
2. Install VirtualBox on the Windows host system.

3. Download the Ubuntu Desktop ISO from ubuntu.com/download/desktop.

Step 2: Create a Virtual Machine

1. Open Oracle VirtualBox Manager.
2. Click New to create a virtual machine.
3. Enter the virtual machine name as Ubuntu Linux.
4. Select Linux as the operating system type and Ubuntu (64-bit) as the version, where available.
5. Select the downloaded Ubuntu ISO image.
6. Configure the required memory, CPU cores, and virtual hard disk.

Step 3: Allocate Resources

An example configuration is shown below. Replace these values with your actual settings.

Resource	Example allocation
CPU	2 virtual CPUs
RAM	4096 MB (4 GB)
Virtual disk	25 GB
Network adapter	NAT
Graphics memory	NA

Resource allocation determines how much of the host computer's capacity the virtual machine can use. The allocated RAM and CPU resources are shared with the host's overall workload, so they should be selected according to the physical system's capabilities.

Step 4: Install Ubuntu Linux

1. Start the virtual machine.
2. Follow the Ubuntu installer instructions.
3. Select the language, keyboard layout, and installation options.
4. Create a user account and password.
5. Complete the installation and restart the virtual machine.

6. Remove or detach the ISO from the virtual optical drive if needed, then boot into the installed Ubuntu system.

Step 5: Verify the Virtual Environment

Open the Ubuntu Terminal and execute these commands.

Check the operating system

cat /etc/os-release

Check CPU information

lscpu

Check allocated memory

free -h

Check disk space

df -h

Check network connectivity

ip addr

Test internet connectivity, if the VM has internet access

ping -c 4 google.com

These commands help verify the guest operating system and inspect the CPU, memory, storage, and network resources visible inside Ubuntu.

6. Resource Allocation and Isolation

CPU allocation

VirtualBox allows the administrator to assign virtual CPUs to Ubuntu. The guest operating system can use these virtual CPUs to execute its processes without requiring a separate physical computer.

Memory allocation

The RAM allocated to Ubuntu is available for running applications and system processes. The host operating system retains access to its own resources, although both systems compete for the physical machine's overall capacity.

Storage allocation

Ubuntu uses a virtual hard disk file stored on the Windows host. The guest operating system treats this virtual disk as its own storage device. The maximum disk capacity depends on the configured virtual disk, while the physical space consumed depends on the disk's allocation type and usage.

Isolation

Ubuntu runs in a separate virtual machine from Windows. Its processes and virtual disk are managed separately from the host operating system. VirtualBox also provides configurable networking and device access. This isolation is useful for testing software and learning Linux without replacing Windows.

7. Demonstration of Isolation

Perform the following tests and record the actual results.

Test	Procedure	Expected observation
Operating system separation	Run cat /etc/os-release in Ubuntu	Ubuntu identifies itself as the guest OS
CPU allocation	Run lscpu	CPU information visible to the guest
Memory allocation	Run free -h	Guest-visible memory information
Storage isolation	Run df -h	Ubuntu's mounted filesystems are displayed
Host and guest separation	Run Ubuntu while Windows remains active	Both operating systems operate simultaneously
Network configuration	Run ip addr	Guest network interfaces are displayed

To demonstrate stronger isolation, create a text file inside Ubuntu and verify that it remains in the guest's virtual filesystem unless you

deliberately share it with the host. Avoid claiming complete security isolation, since shared folders, clipboard integration, networking, and other VirtualBox settings can create communication paths between the host and guest.

8. Results

The Ubuntu Linux virtual machine was configured and executed successfully using Oracle VirtualBox on a Windows host system.

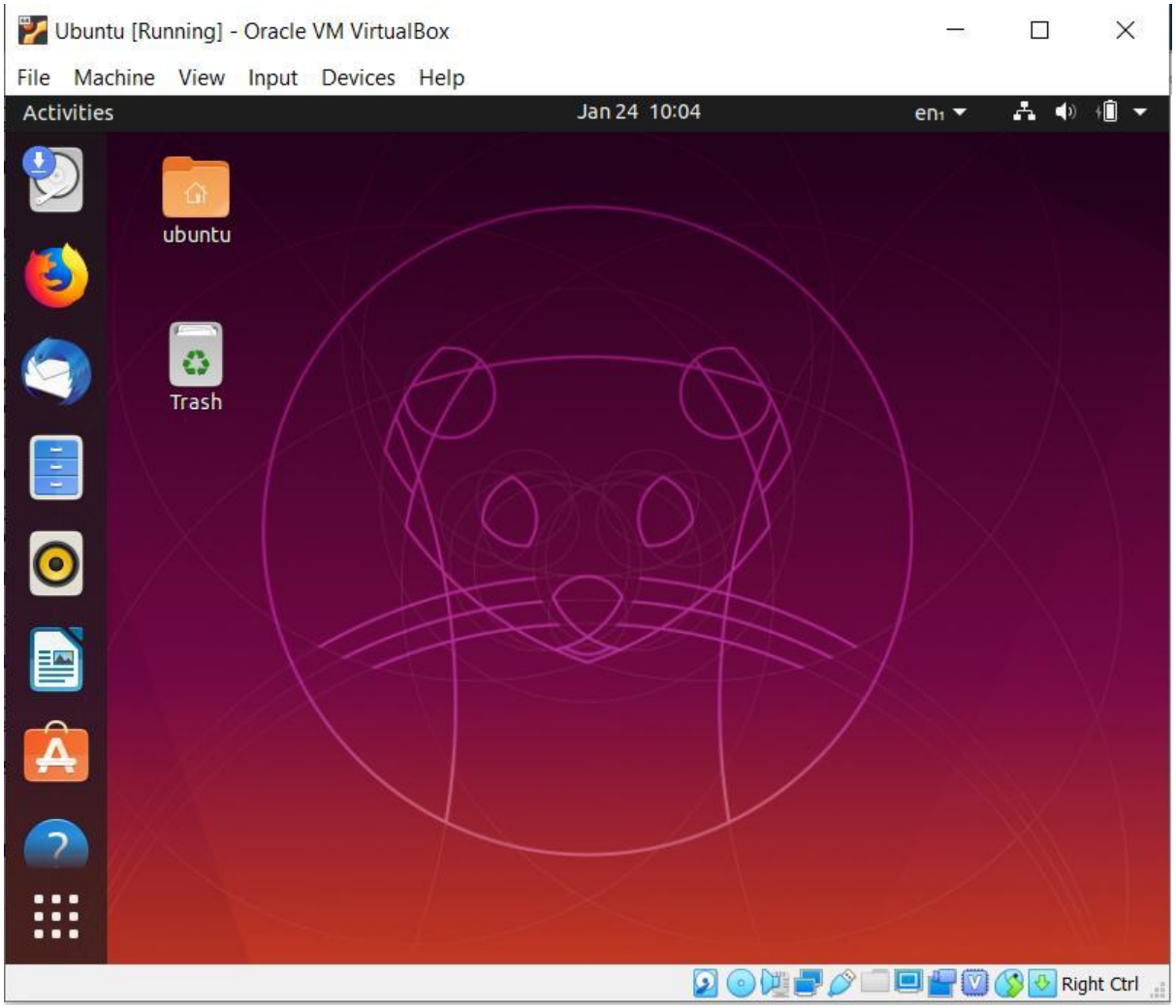
The activity demonstrated the allocation of virtual CPU, RAM, and storage resources. Ubuntu operated as a guest operating system while Windows remained active as the host. System commands were used to inspect the virtual machine's CPU, memory, disk, and network configuration.

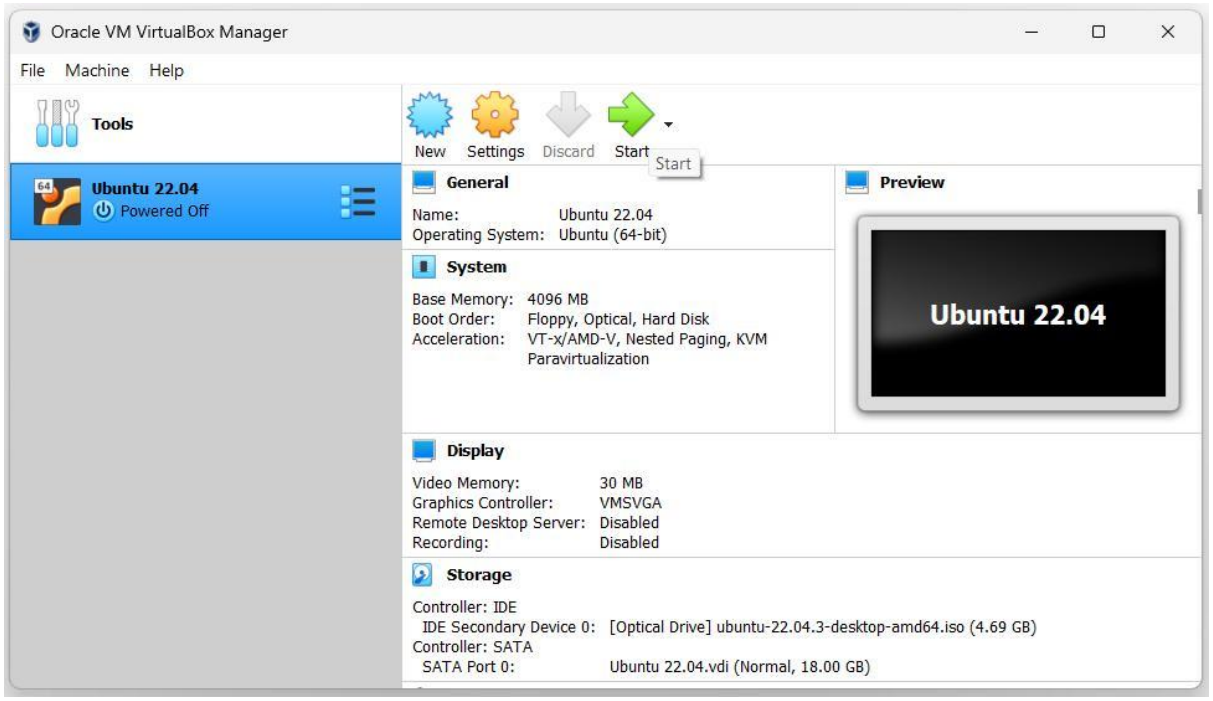
Write that the implementation was successful only after verifying it on your system.

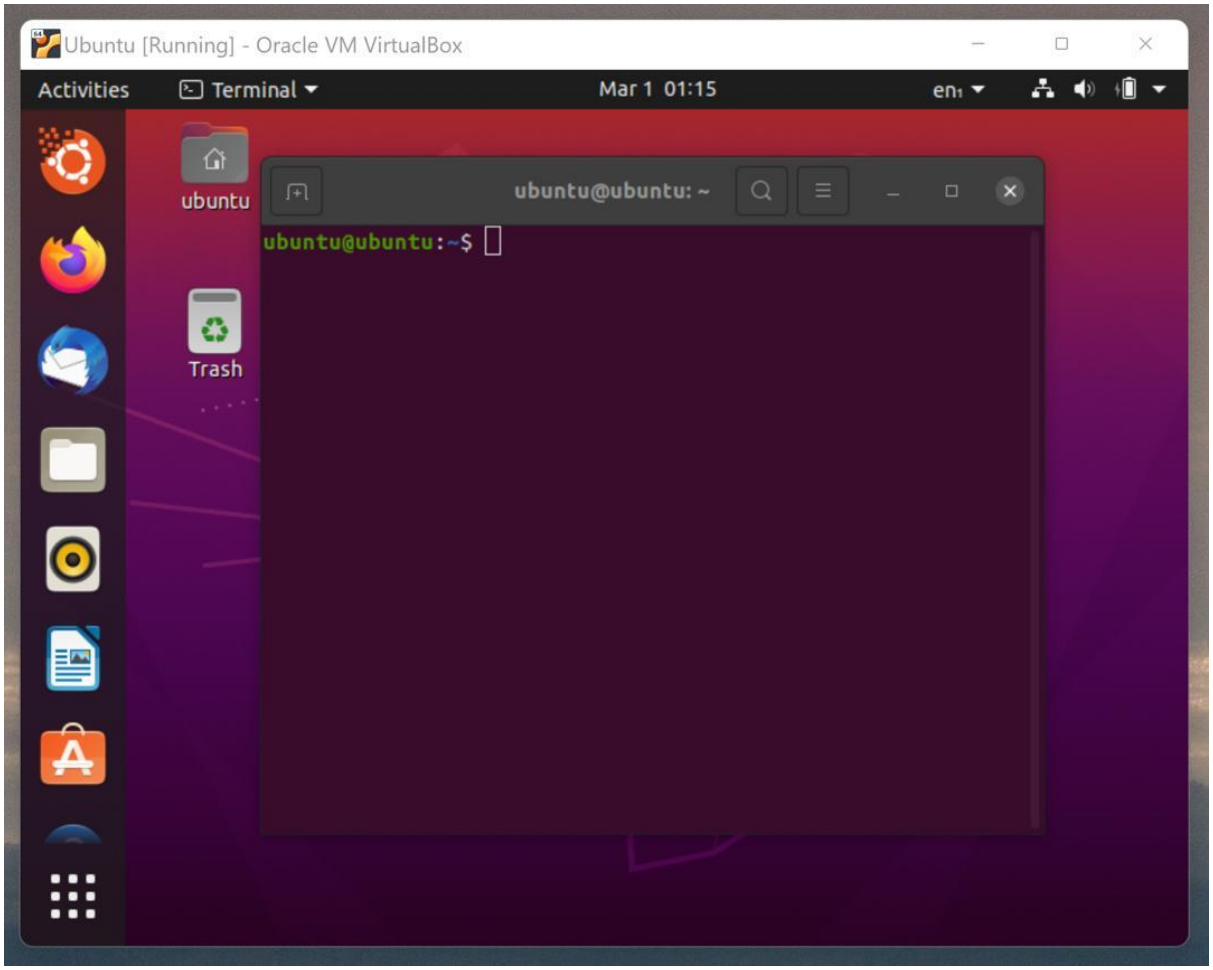
9. Advantages of Virtualization

1. Multiple operating systems can run on one physical computer.
2. It reduces the need for separate physical hardware.
3. It provides an environment for software development and testing.
4. Virtual machines can be configured with specific CPU, RAM, and storage limits.
5. Snapshots can help restore a virtual machine to an earlier state.
6. It provides useful separation between guest and host environments.

10. Screenshots


Ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Activities
Jan 24 10:04
en
ubuntu
Trash
Right Ctrl


Oracle VM VirtualBox Manager
File Machine Help
Tools
Ubuntu 22.04
Powered Off
New
Settings
Discard
Start
General
Name: Ubuntu 22.04
Operating System: Ubuntu (64-bit)
System
Base Memory: 4096 MB
Boot Order: Floppy, Optical, Hard Disk
Acceleration: VT-x/AMD-V, Nested Paging, KVM
Paravirtualization
Preview
Ubuntu 22.04
Display
Video Memory: 30 MB
Graphics Controller: VMSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: IDE
IDE Secondary Device 0: [Optical Drive] ubuntu-22.04.3-desktop-amd64.iso (4.69 GB)
Controller: SATA
SATA Port 0: Ubuntu 22.04.vdi (Normal, 18.00 GB)



11. Conclusion

This activity provided practical knowledge of virtualization using Oracle VirtualBox. Ubuntu Linux was installed as a guest operating system on a Windows host machine. CPU, memory, storage, and network settings were configured and examined. The experiment demonstrated how virtualization enables multiple operating systems to operate on the same physical computer while maintaining separate guest environments.